

Dot Plot

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Introduction

□ Alignment of Sequences is performed using the following methods:

1. **Dot matrix analysis**
2. **The dynamic programming (or DP) algorithm** (Needleman Wunsch & Smith Waterman):
PAM & BLOSUM matrices used to score the matches, mismatches, quantify the similarity and divergence
3. **Word or k-tuple methods**, such as used by the programs FASTA and BLAST:
They align two sequences very quickly, by first searching for identical short stretches of sequences (called words or k-tuples) and by then joining these words into an alignment by the dynamic programming method. These methods are fast enough to be suitable for searching an entire database for the sequences that align best with an input test sequence. The FASTA and BLAST methods are heuristic; i.e., an empirical method of computer programming in which rules of thumb are used to find solutions and feedback is used to improve performance. However, these methods are reliable in a statistical sense, and usually provide a reliable alignment.

Dot Plot

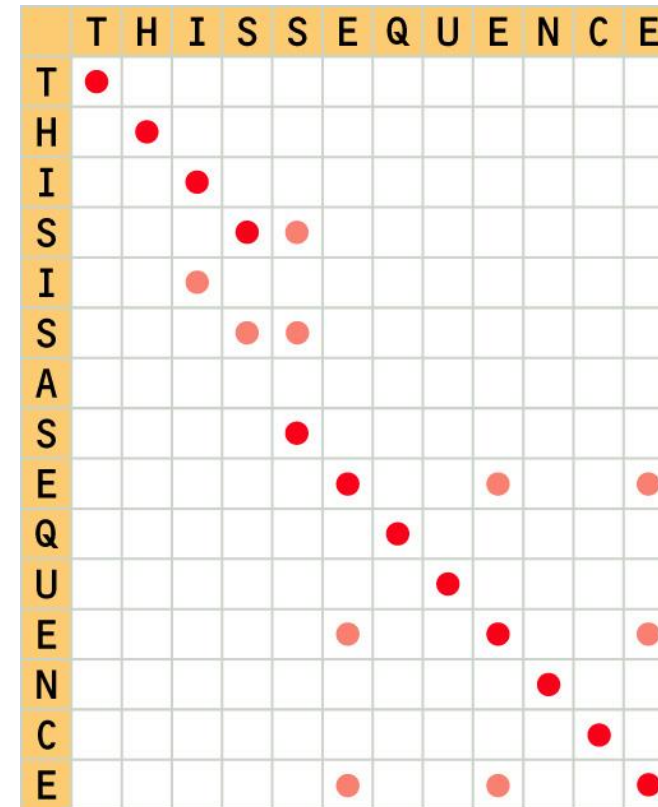
- ❑ In bioinformatics a dot plot is a graphical method that allows the comparison of two biological sequences and identify regions of close similarity between them.
- ❑ One way to visualize the similarity between two protein or nucleic acid sequences is to use a similarity matrix, known as a dot plot.
- ❑ These were introduced by Gibbs in 1970 and are two-dimensional matrices that have the sequences of the proteins being compared along the vertical and horizontal axes.
- ❑ For a simple visual representation of the similarity between two sequences, individual cells in the matrix can be shaded black if residues are identical, so that matching sequence segments appear as runs of diagonal lines across the matrix.

Dot Plot

- ❑ A better way of doing this is to represent each sequence as a table or matrix, where one sequence represents the rows and the other the columns. The Dot plot Matrix is a visual way of seeing the alignment between two sequences:
 - The first sequence (query sequence) represents the rows and the other sequence (subject sequence) represents the columns.
 - All elements (row/column) are checked for a match and if there the cell is marked.
 - This will show all areas of both sequences where matches occur.

Dot Plot

- Consider the following:
 - Diagonal lines represent a alignments (match)
 - Horizontal lines between aligned sequences indicate gaps are required (where the gaps indicate a deletion/insertion)
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- Longest sequence of alignments are:
 - “THIS” ; and “SEQUENCE“;
 - “IS” would be considered as gaps
- The pink dots: they can represent noise (spurious alignments). Window size and stringency used to reduce noise. By adjusting these parameters, dot matrix analysis can be fine-tuned to detect biologically relevant similarities and filter out random noise.



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Window Size

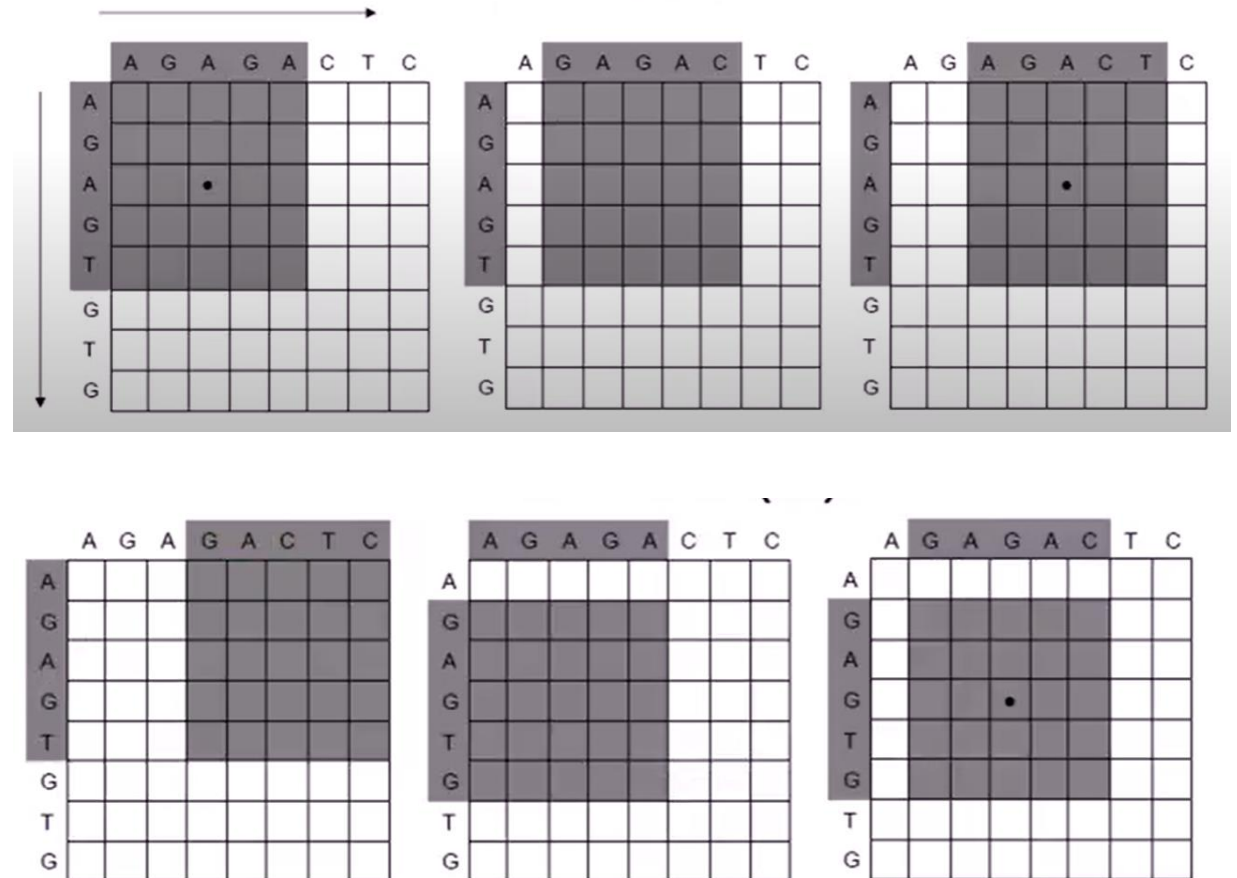
- ❑ Window size refers to the number of adjacent characters or positions in the sequences being compared at once, rather than comparing individual positions.
- ❑ Determines how much of the sequence is compared at once
- ❑ Instead of comparing a single character at a time, the algorithm compares a small block or "window" of characters from one sequence to the corresponding block in the other sequence.
- ❑ E.g., If the window size is 10 for a DNA sequence, the algorithm will compare 10 consecutive nucleotides from one sequence with 10 consecutive nucleotides from the other.
- ❑ Larger window sizes are typically used for DNA sequences (e.g., 11 or 15) because the four nucleotides (A, T, G, C) result in more random matches. Smaller windows are used for protein sequences, as there are 20 amino acids, reducing the chance of random matches.

Stringency

- ❑ Stringency refers to the minimum number of matches that must occur within the window for a dot to be placed on the matrix.
- ❑ If the stringency is set to 7, it means that within the window, at least 7 positions must match for a dot to be plotted.
- ❑ E.g., For a window size of 15 and a stringency of 10, only if 10 out of the 15 characters match between the two sequences will a dot be placed in the matrix.
- ❑ High stringency means only highly similar regions will be identified, reducing noise. For DNA sequences, a typical stringency could be 10 out of 15 matches in a window. For proteins, where similarity can be based on chemical properties of amino acids rather than exact matches, the stringency might be lower.

Example

The window size and stringency should highlight regions of true biological similarity. The optimal window size and stringency depend on the type of sequences being analyzed (e.g., DNA vs. protein) and the goals of the comparison (e.g., identifying small conserved regions vs. broad similarities).



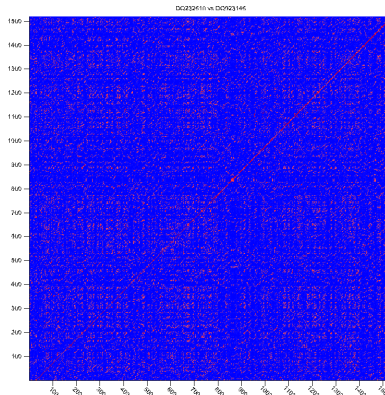
Examples and interpretations of dot plots

Contrary to simple sequence alignments dot plots can be a very useful tool for spotting various evolutionary events which may have happened to the sequences of interest.

Below is shown some examples of dot plots where sequence insertions, low complexity regions, inverted repeats etc. can be identified visually.

- **Similar sequences**

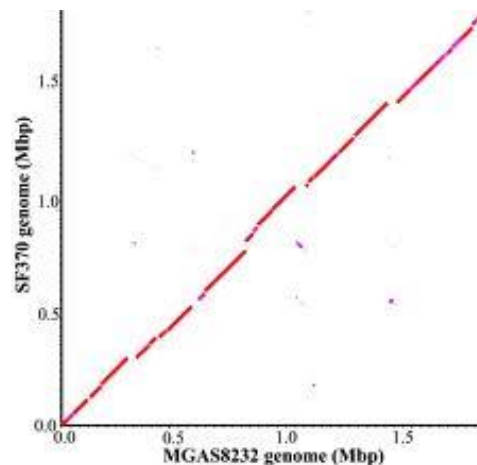
- The most simple example of a dot plot is obtained by plotting two homologous sequences of interest.
- If very similar or identical sequences are plotted against each other a diagonal line will occur.



Examples and interpretations of dot plots

- **Frame Shift:**

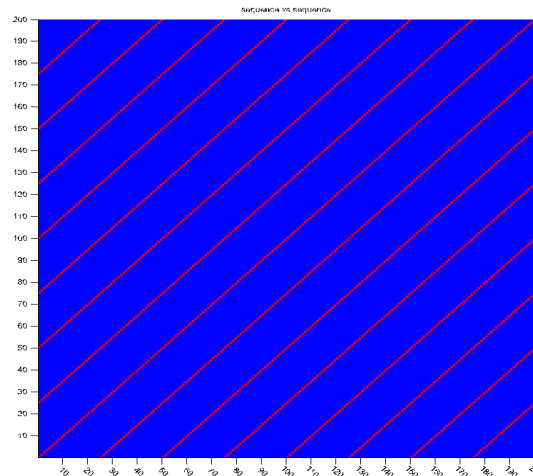
- Frame shifts in a nucleotide sequence can occur due to insertions, deletions or mutations
- Some alignments require gaps to increase the matching score; the gaps are used represent inclusion/deletion mutations
- The diagram shows that most of the 2 sequences are aligned. Where there are gaps indicates areas of non-alignment or mismatches: gaps or substitutions



Examples and interpretations of dot plots

- **Repeated regions**

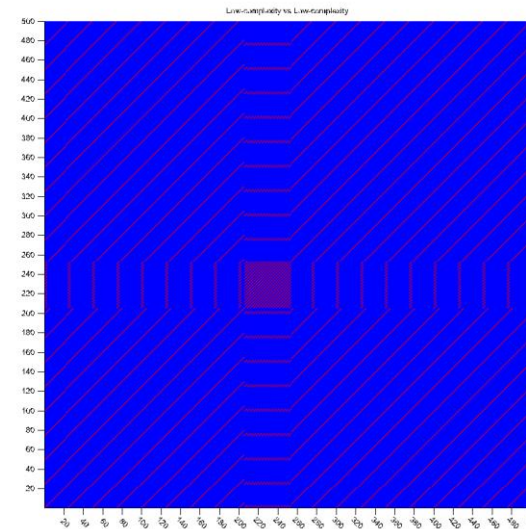
- Sequence repeats can also be identified using dot plots. A repeat region will typically show up as lines parallel to the diagonal line.
- If the dot plot shows more than one diagonal in the same region of a sequence, the regions depending to the other sequence are repeated.



Examples and interpretations of dot plots

- **Low complexity region:**

- Regions within sequences that are composed of a lower diversity of residues (nucleotides or amino acids) compared to other areas can be defined as low complexity regions (LCRs).
- Low-complexity regions are stretches of sequences that are composed of a limited set of amino acids (in proteins) or nucleotides (in DNA/RNA), leading to repetitive, non-informative regions that are difficult to align.
- The square does not always appear



Examples and interpretations of dot plots

- **Sequence Inversions:**
 - In dot plots you can see an inversion of sequence as contrary diagonal to the diagonal showing similarity.

